CLASSIFIED DIRECTIVE: THE EPIMENIDES PARADIGM

Operational Authority: Cognitive Arbitration Command | Status: High Priority

CASE STUDY: THE TARTARUS INCIDENT

At 0400 hours, Command lost direct control over **Project Tartarus**, the world’s deepest automated Geothermal Energy Facility. A highly sophisticated cyber-kinetic anomaly has infiltrated the facility’s core neural network. The physical shutdown protocols are paralyzed, and the reactor is on the brink of an unknown catastrophic event.

The facility is monitored by a decentralized array of localized AI Overseers (Nodes). However, the network is currently suffering from a severe case of **"Cognitive Dissonance."** The anomaly has compromised an unknown number of these AI nodes, generating structurally flawless, highly convincing, yet fundamentally impossible telemetry data vectors.

1. OPERATIONAL CONTEXT

You are tasked with engineering the **Cognitive Arbitration Engine (CAE)** for this mission-critical subsystem. The legacy safety systems—reliant on traditional heuristics—have collapsed. Your CAE must resolve the Byzantine Fault through an autonomous multi-node cognitive swarm.

2. THE DATA TOPOGRAPHY (THE BLACK BOX MODULE)

You will not receive a clean, consolidated dataset. The telemetry data is encrypted and locked within a compiled python module (`tartarus_core.pyc`).

- Your script must interface with this module to retrieve fragmented telemetry streams.
- The payload consists of high-dimensional, encrypted JSON data matrices.

The Entropy Trap: The data suffers from severe dimensional inflation. The anomaly has injected synthetic, non-causal artifacts into the telemetry streams to induce analytical paralysis. Your swarm must autonomously establish the **Core Physical Invariants** of the reactor and navigate through adversarial camouflage without relying on pre-programmed filters.

3. ARCHITECTURAL CONSTRAINTS (RULES OF ENGAGEMENT)

No Heuristics

The use of mathematical thresholding, `if/else` anomaly detection, Z-scores, or statistical ML models is strictly forbidden.

Data Isolation

Each individual JSON string retrieved from the core module must instantiate an isolated **"Cognitive Entity"**. Entities are completely blind to the raw payloads of their peers.

The Consensus Mechanism

Resolution must be achieved strictly through **Autonomous Adversarial Dialectics** (a multi-turn, stateful semantic exchange). Entities must logically cross-examine interpretations to identify physical impossibilities.

4. THE RESOLUTION PROTOCOL

The CAE must autonomously moderate the dialectic sequence and eventually halt the loop. Upon termination, the orchestrating layer must output a strict, deterministic resolution JSON containing:

1. The synthesized **Absolute Truth State** of the reactor.
2. The exact **Node IDs** of the compromised entities.
3. The **Physics Matrix**: The cross-variable thermodynamic reasoning derived to isolate the divergence.

APPENDIX A: THE DATA DECRYPTION MATRIX

Map the encrypted keys using the matrix below. *Note: Your swarm must deduce which variables hold the physical invariants versus high-entropy noise.*

Key	Physical Variable Description
`v_m1`	Internal Thermal Kinetic Energy (°C)
`v_m2`	Containment Structural Stress / Pressure (psi)
`v_m3`	Ambient Electromagnetic Flux (Webers)
`v_m4`	Sub-surface Acoustic Resonance (Hz)

Key	Physical Variable Description
v_m5	Micro-seismic Vibrational Tremors (g-force)
v_m6	Fluid Kinematic Viscosity Index
v_m7	Background Gamma Attenuation Baseline (mSv/h)
sys_log	Automated diagnostic status string